

Generated Data, Real Consequences: Why LLMs Are Inappropriate for Development Aid Targeting

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Abstract

Large language models (LLMs) can match demographic group means in opinion survey contexts, prompting interest in their use for generating synthetic household data in development economics. We evaluate the empirical case and the normative case for that proposal across 17 nationally representative household surveys (131,906 households) drawn from the Living Standards Measurement Study program and comparable country-specific surveys. Empirically, we document a specific pattern of country-level stereotyping. When an LLM is asked to answer household survey questions in the persona of a specific household, its downstream consumption predictions concentrate on country identity: at individual-persona prompts (L3), country identity explains 0.51 of the variance in the LLM’s predictions against 0.12 in true household consumption, a stereotyping multiple of 4.4. A variance decomposition of the LLM’s own predictions gives the same finding directly: country identity holds between 42-56% of that variance as household detail included in the prompt increases. Using the same pipeline on 10,000 US households from the American Community Survey and substituting state for country, we find a different and more expected pattern: state identity holds 62% at L1 and then falls to 8%, 2.6% and 2.2%, and the stereotyping multiple stays at or below 1.4 at every level. Fine-tuning on empirical survey distributions reduces marginal distributional divergence by 36% on average, lifts household-level rank correlation with true consumption in every evaluable country, and lowers the country stereotyping to at or below the level in the surveys themselves. Targeting does not follow: exclusion error falls five percentage points, and in only half the countries, because the fine-tuned model already sits near the information ceiling of its inputs. Normatively, we argue that even absent the stereotyping, LLM-generated household data would remain disqualifying for allocative decisions about specific households, because its group-level inferences are not grounded in observations of the population the decisions affect, an evidentiary structure inappropriate for high-stakes allocation.

1 Introduction

Recent work demonstrates that large language models (LLMs) can match group-level means in US public opinion polling contexts, and can even match group-level distributions after fine-tuning (Argyle et al. 2023; Bisbee et al. 2024; Suh et al. 2025). This development has raised hopes for broader applications in social science research. Develop-

ment economics appears a natural extension, since household surveys comprise the primary data infrastructure for poverty measurement and program targeting. Those surveys are expensive. The Living Standards Measurement Study (LSMS) is a program of separate nationally representative surveys, coordinated by the World Bank; a single such survey round costs US\$830k–2.5 million, and the World Bank has committed to 390 multi-topic household surveys of this kind before 2030 (World Bank 2023b). If LLMs could generate realistic household data from minimal demographic information, the cost savings would be transformative. The demand this would meet is institutional and active. The World Bank’s 2025 Poverty Prediction Challenge, which drew 1,322 registered participants, was motivated by the fact that consumption surveys “are conducted infrequently due to cost and logistical constraints” (World Bank 2025), and survey-to-survey imputation is already an accepted substitute where consumption is unobserved (Dang et al. 2025). Targeting has consistently used novel technologies as they arrive to better identify and assist those in need, including satellite data, mobile phone data, and social media data (Chi et al. 2022). In the adjacent survey industry the substitution from human to silicon has already happened: in a 2024 survey of over 3,000 market researchers, 71% expected synthetic responses to constitute more than half of data collection within three years (Qualtrics 2025). The question this paper answers is therefore not whether anyone has announced a plan to replace survey rounds with an LLM, but whether such a plan would induce bias and harm, who would experience those harms, and whether those harms can be justified when compared against alternative methods.

The case for testing LLMs in this role is not obviously wrong. Pretraining on web-scale corpora leaves large language models with substantial factual knowledge about the world that they can retrieve at inference time (Petroni et al. 2019; Roberts, Raffel, and Shazeer 2020). The plausibly relevant material for development targeting has two sources. One is outside accounts: development reports, academic literature, ethnographic studies, and general documentation of how demographic characteristics correlate with consumption in low- and middle-income settings. The other is direct digital traces from the populations themselves. Mobile phone and internet access in sub-Saharan Africa and South Asia have grown substantially in the decades since Aker

and Mbiti (2010) documented the early penetration wave, and the resulting digital exhaust could in principle provide LLMs with first-hand material about the same populations our surveys sample. A country-specific supervised model is trained on a comparatively small sample (2,500 households in our setup) and cannot access either kind of material directly; an LLM prompted with the same demographic inputs could in principle bring both to bear, and produce a better conditional expectation. Whether that potential materializes depends on two further questions: whether digital traces from these populations are ingested into pretraining at the rate traces from Global North populations are, and whether the qualitative material of either channel supports a workable mapping from demographic description to consumption. This paper answers the empirical version of those questions.

An aggregate accuracy shortfall for populations under-represented in web text is by now a well-established finding (Santurkar et al. 2023; Bender et al. 2021; Naous, Ryan, and Ritter 2024), so a performance gap between a country in the Global North and South on our task is a reasonable prior. What this paper adds is a comprehensive characterization of the shape of that shortfall: how it varies with prompt granularity across 17 countries, who bears its costs across demographic groups within those countries, and what mechanisms might produce it. On the last point we provide a partial diagnosis. The internals of modern models are not fully inspectable, and we do not claim to definitively adjudicate why the documented behavior arises. The two candidate mechanisms we discuss in Section 4.1 are each consistent with the pattern we observe and with our activation-probing results.

While LLMs have shown promising results in opinion polling contexts, development targeting differs fundamentally in both stakes and structure. Public opinion research aggregates views across demographic groups to estimate population-level sentiment; development targeting allocates scarce benefits to geographic areas or specific households, where errors have direct welfare consequences. This distinction also carries normative weight: the legitimacy of algorithmic allocation decisions depends on the evidentiary adequacy of their inputs (Porter 1996).

This paper argues that LLMs are inappropriate as a substitute for household survey data in poverty targeting, on three grounds: the LLM’s synthetic survey responses concentrate on country-typical values rather than discriminating among households within a country; this concentration magnifies exclusion errors across the population and redistributes them from smaller households onto larger ones. Even if this were not the case, the qualitative material an LLM is built on does not support contestability, accountability, or representativity in the way a proxy built on observed household data does. The mechanism is training-text driven rather than prompt-structure driven: the LLM defaults to country-typical values when its training text underrepresents the population under study, and a replication on 10,000 US households from the American Community Survey, substituting state for country, inverts the pattern. Section 4.3 reports the magnitudes.

The interventions usually proposed to improve LLM performance do not resolve this problem. Fine-tuning on em-

pirical survey distributions, following Suh et al. (2025), reduces the LLM’s marginal distributional divergence by 36% on average, lifts household-level rank correlation in every evaluable country, and lowers the country stereotyping to at or below the level in the surveys themselves. Exclusion error falls by five percentage points, and in only half the countries. Fine-tuning teaches the model correct group-level marginals; within-cell discrimination requires information the demographic prompt does not contain, and no amount of fine-tuning supplies it. Scaling the model does not close the gap either: what the training text encodes about a country’s poor households does not change with capability, and neither the LLM’s next-token output, its fine-tuned variant, nor a linear read-out of its internal representations recovers within-cell signal on our data. Section 5 develops this for fine-tuning; Section 6.1 does so for a linear ridge probe on the model’s internal representations, which shows the collapse is present in the model’s representations themselves.

We further argue that even were this shortfall to close, LLM-generated data would remain disqualifying for allocative decisions about specific households, for reasons that do not depend on its current accuracy: its inferences are group-level inferences not grounded in observations of the population the decision affects, drawn from a corpus that is undisclosed and, on its publicly characterized slice, represents these populations sparsely and indirectly. The proxy means tests (PMTs) we use as the foil throughout the paper are imperfect on this dimension too; the normative argument rests on the relative difference. Section 6.2 develops this point.

We test the empirical claims by evaluating four prompting strategies against 131,906 households drawn from 17 separate nationally representative household surveys, one per country, spanning sub-Saharan Africa, South Asia, and Latin America. The underlying surveys come from the World Bank’s Living Standards Measurement Study (LSMS) and from country-specific household survey systems (India’s NSS, Colombia’s SISBEN, Kenya’s KCHS, among others), harmonized in Sahoo et al. (2025). The design is deliberately cross-country so that no one survey drives the empirical conclusions. Our prompting strategies correspond to concrete policy deployment scenarios. L1 and L2 are the data-sparse cases. L1 represents purely geographic targeting: only rough administrative geography is available, analogous to a poverty map, while L2 adds age bins for the household head. L3 represents the census or registry case: when a country has demographic data on individuals (age, gender, occupation, region) from a prior census or registration system but no current consumption survey. L4 represents the data-rich case: when comprehensive household-level data are available, making the question of LLM utility stark, since an organization with L4-level data has little need for synthetic data generation in the first place. For each scenario, we compare an LLM-PMT to a level-matched True-PMT trained on the same input data: does routing available data through an LLM add value over fitting a PMT to it directly?

In each scenario, the LLM functions as a survey imputation tool: it generates synthetic responses for real households or geographic cells with known characteristics, which

are then used as inputs to a PMT. The targeting exercise evaluates whether those imputed responses support better allocation decisions about those real households. We evaluate it this way because real-world targeting systems require auditable, observable household characteristics to justify allocation decisions and allow households to contest them; direct black-box poverty prediction from demographics fails this transparency requirement.¹

2 Related Work

2.1 LLMs as Survey Simulants

A growing literature explores LLMs as stand-ins for human survey respondents. Argyle et al. (2023) demonstrate that GPT-3 can reproduce subgroup-level opinion distributions across demographic strata when prompted with persona descriptions, a technique they term “silicon sampling.” Bisbee et al. (2024) find that LLMs match group means for US political attitudes and propose avatar prompting as a survey methodology. Horton, Filippas, and Manning (2023) extends this to economic preferences, showing that *homo silicus* replicates standard experimental findings in labor and consumer economics. Park et al. (2023) develop generative agents capable of simulating social behavior at the individual level, though in synthetic rather than real-world calibration settings.

These results share a common structure: LLMs perform well at reproducing aggregate patterns for well-represented populations and low-stakes applications. The opinion polling context is relatively forgiving: errors average out across respondents, the target population (US adults) is heavily represented in training data, and the consequences of error are limited. Development targeting differs on all three dimensions.

Suh et al. (2025) most directly motivate fine-tuning as an improvement strategy, showing that models fine-tuned on empirical opinion distributions substantially reduce divergence from ground truth distributions. We adapt this approach and find similar distributional improvements, but show that distributional accuracy does not reliably translate to individual-level targeting accuracy.

2.2 Development Targeting Methods

Because household consumption is expensive to measure directly, most large-scale poverty targeting programs do not measure it at every eligibility decision. They substitute a cheaper set of observable proxies for consumption itself. That substitution is the proxy means test (Grosh, Baker et al. 1995), and it has a long technical literature behind it (Coady, Grosh, and Hoddinott 2004). Concretely, a PMT is a supervised model such as a regularized linear regression or a tree ensemble (McBride and Nichols 2018) fitted on a training sample in which both proxy variables and a welfare metric

of interest such as consumption are jointly observed. Typical proxy variables are easily observable measurements of living standards: asset ownership, housing quality, demographic composition of the household, and geography. Once fitted, the model produces a predicted consumption score for households on which only the proxies are collected; program eligibility is then set by ranking that score. PMTs are now standard in social protection programs across sub-Saharan Africa, South Asia, and Latin America (Gentilini et al. 2019). In this paper the PMT plays two roles: as the level-matched empirical baseline and as the normative foil to the LLM route. For any candidate method that claims to impute survey responses from cheaper inputs, a PMT trained on exactly those cheaper inputs asks whether routing them through the imputation adds value over fitting a supervised model to them directly. This is the comparison we run at each of our four granularity levels.

2.3 Synthetic Data for Development

The idea of synthetic survey data is not new, but LLMs represent a qualitatively different approach. Multiple imputation fits a model on real observations of the relevant population and uses that model to generate plausible values for missing data, propagating the resulting uncertainty into downstream inference (Rubin 1993); the model is explicit and the inference is auditable. LLM generation has none of these properties: the training corpus is undisclosed (Bommasani et al. 2023), public characterizations indicate it overrepresents populations outside those under study (Bender et al. 2021; Dodge et al. 2021), and the generative process produces confident-looking responses without an honest representation of their uncertainty (Xiong et al. 2024). In a closely related demonstration outside the development setting, Westwood (2025) reports that LLM-generated survey responses pass a comprehensive battery of standard quality checks at 99.8% and conform systematically to the researcher’s inferred hypothesis, producing internally coherent data whose biases are not detectable from the responses themselves. The use of LLMs to generate quantitative survey microdata for poverty measurement is, to our knowledge, untested at scale prior to this work.

2.4 Accountability in Algorithmic Targeting

The use of algorithmic systems in public benefit allocation has attracted substantial critique (Eubanks 2018; Benjamin 2023). A growing literature examines the specific conditions under which algorithmic targeting is legitimate (Barocas, Hardt, and Narayanan 2023). Porter (1996) argues that quantitative methods in governance create a form of accountability through their apparent objectivity, but can obscure the assumptions and exclusions embedded in measurement choices. This argument applies with particular force to LLM-generated targeting data: the impressive technical performance of LLMs in benchmark contexts provides no guarantee of adequacy in development targeting, and the opacity of the generative process makes errors difficult to contest or remedy.

¹In auxiliary experiments we also tested a direct-prediction design, in which the LLM received the same demographic persona but was asked to report the household’s daily per-capita consumption directly. Performance was worse than the imputation design reported here; the transparency objection is binding regardless of relative empirical performance.

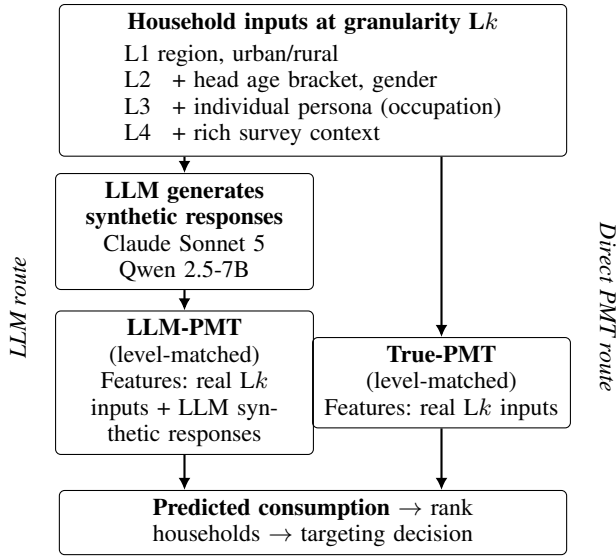


Figure 1: Experimental design. The pipeline is instantiated separately for each of the 17 countries in our sample and for each granularity level L1–L4. At each level, the same household inputs are routed two ways. On the LLM route, an LLM (Claude Sonnet 5 or Qwen 2.5-7B) is prompted with the L_k inputs and generates synthetic responses to the country’s survey questions, which are then passed to a country-specific, level-matched PMT trained on real survey data. On the direct PMT route, the same estimator is applied directly to the real answers to the same survey questions plus the same real L_k inputs, without an LLM in the loop. The two PMTs are separately fitted models that use the same estimator and the same training households on the same feature set; the LLM route uses LLM-synthesized answers to the survey questions while the direct PMT route uses actual answers. Both paths produce a predicted consumption used to rank households and set eligibility under a fixed budget.

3 Data and Methods

3.1 Survey Data

Our household survey data come from the Living Standards Measurement Study (LSMS) program coordinated by the World Bank as well as country-specific household survey programs. These surveys collect detailed household-level data on consumption expenditure, assets, housing quality, education, agricultural production, and demographic composition. The primary outcome variable in our analysis is per capita consumption expenditure, expressed in 2017 PPP USD per person per day, the standard measure used to define the \$2.15/day extreme poverty threshold (World Bank 2023a).

We use data from 17 countries spanning sub-Saharan Africa, South Asia, and Latin America; per-country sample sizes, survey years, and welfare summaries are reported in Appendix Table A2. Countries were selected to maximize geographic and economic diversity while using surveys with consumption data of sufficient quality to train PMTs.

For each country, we construct two samples. The PMT

training sample consists of 2,500 households drawn via stratified proportional sampling across administrative regions, used to train per-country PMTs. The evaluation sample consists of the remaining households, used to evaluate the performance of PMTs trained on LLM-generated responses.

3.2 Prompting Design and Granularity Levels

We test four prompting strategies of increasing data richness, designed to span the practical range of LLM deployment scenarios:

L1 (Coarsest): One prompt per geographic cell (ADM-1 region $\times$ urban/rural classification). The LLM is asked to respond as a representative household in that cell, without individual-level information. This corresponds to a coarse poverty map at the ADM-1 $\times$ urban/rural resolution, and represents the scenario where no individual household data are available.

L2 (Coarse): One prompt per demographic cell (ADM-1 $\times$ urban/rural $\times$ household head age bracket $\times$ gender). Finer demographic granularity, but still no individual household identity.

L3 (Standard): One prompt per individual household, providing a full persona: region, urban/rural, head age, head gender, and occupation. This is the avatar prompting approach of Bisbee et al. (2024) adapted to the development context. This is based on a plausible deployment scenario, requiring only the minimal demographic data typically available from administrative registries.

L4 (Rich): One prompt per individual household, augmented with all available survey variables as context: asset ownership, housing characteristics, agricultural production, household composition, and education levels. This upper-bound scenario asks whether comprehensive household data, if available, would enable adequate targeting.

Prompts ask the LLM to respond in tab-separated format to a standardized set of survey questions adapted from the target country’s survey instrument (see appendix for full prompt templates). We use Claude Sonnet 5 as the primary generation model, and Qwen2.5-7B-Instruct at 4-bit quantization as the open-weights companion. All Claude Sonnet 5 responses were collected via the Anthropic Batch API between 2026-09-06 and 2026-09-07.

The choice of Sonnet 5 reflects our aim to characterize what a strong current-generation model can do on this task. Public human-preference leaderboards (Chiang et al. 2024) placed Sonnet 5 among the highest-scoring publicly available systems at the time of data collection. The choice of Qwen2.5-7B-Instruct reflects different constraints: the activation-probing exercise reported in Section 6.1 and the fine-tuning experiments reported in Section 5 both require access to internal representations and adaptable weights, so they can only be run on an open-weights model. Qwen2.5-7B-Instruct was among the highest-scoring open-weights instruction-tuned models at the 7B parameter scale at the time of writing, and 4-bit quantization allows inference on a single consumer-grade GPU (approximately 3.5 GB VRAM). Running both a closed-provider model and a

smaller open model tests whether the patterns we document are specific to a particular provider.

3.3 PMT Model Construction

For each country and granularity level, we train a level-appropriate PMT model with log per capita consumption as the target variable. For L3 and L4, we use Random Forest with 10-fold cross-validated hyperparameter selection; input features are selected from permissible survey variables by ranking Random Forest feature importances from an initial full-feature model, with four default features (region, second administrative unit, urban/rural, household size) always included and the top N additional features by importance retained (12–15 total, country-specific).

For L1, we train a geographic PMT using region and urban/rural status only. For L2, we train a coarse demographic PMT using region, urban/rural, head age bracket, and head gender. Both L1 and L2 models use a Lasso regression (Tibshirani 1996) with 10-fold cross-validation. For L3, we construct a standard individual-level PMT (12–15 features selected by RF feature importance). For L4, we train an expanded PMT using all available survey variables without feature selection. In each case the PMT uses exactly the information available to the LLM at that level, ensuring PMT performance is a matched upper bound. The choice of estimator at each level reflects standard PMT practice: regularized linear models when the feature set is low-dimensional and categorical, and tree ensembles when it is high-dimensional with mixed types and interactions (McBride and Nichols 2018).

For each level of granularity, L1-L4, there are two PMTs. One uses only the information provided to the LLM in the prompt. The other uses this information in addition to the LLM-generated responses.

3.4 Evaluation Metrics

We evaluate performance along two dimensions that correspond to the two use cases in the literature: (1) distributional accuracy, appropriate for population-level inference; and (2) targeting accuracy, appropriate for both cell- and household-level allocation decisions.

Rank correlation. Spearman’s ρ between PMT predictions derived from LLM-generated data and true cell-mean consumption (L1/L2) or individual consumption (L3/L4). Measures whether LLM-generated data enables correct ordering of geographic cells (L1/L2) or individual households (L3/L4) by poverty level.

Targeting coverage. Given a fixed budget equal to the country-specific poverty rate ($\bar{p}_c$ from World Bank PIP (World Bank 2023a)), households are ranked by predicted consumption and the poorest fraction $\bar{p}_c$ are “targeted.” Coverage is the share of truly poor households reached: $\text{cov} = \text{TP}/N_{\text{poor}}$.

Exclusion error. $\text{excl} = \text{FN}/N_{\text{poor}}$, the share of truly poor households who are wrongly excluded from benefits. Note that $\text{excl} = 1 - \text{cov}$.

Coverage Δ . $\Delta = \text{coverage}_{\text{PMT}} - \text{coverage}_{\text{LLM}}$, the difference in poor-household coverage between a fully observed PMT and the LLM at the same budget. This is rep-

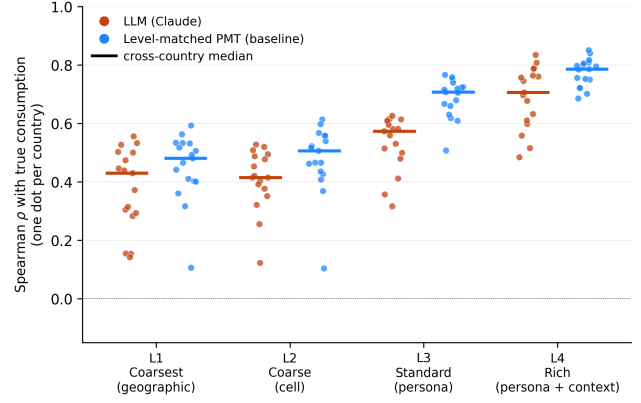


Figure 2: Spearman rank correlation with true per-capita consumption at each granularity level, one dot per country. Rust dots (left of each pair) are the LLM route (Claude Sonnet 5); blue dots (right) are the level-matched PMT baseline fit directly on the same inputs. The horizontal bar is the cross-country median.

resentative of, without being a measurement of, the welfare cost of using LLM-generated data.

Poverty lines are set at \$2.15/day (2017 PPP) per person per day, consistent with the World Bank extreme poverty threshold. Budget fractions use country-specific poverty rates from the World Bank PIP database to avoid artificially inflating coverage metrics for high-poverty countries.

4 Results

Mean rank correlations between the LLM-PMT and true consumption sit in the 0.38–0.69 range across the four granularity levels (Figure 2): 0.38 at L1, 0.41 at L2, 0.53 at L3, 0.69 at L4, with substantial country-level variation. Values in this range would ordinarily be taken as evidence that LLM-generated data carries meaningful signal about which households are poorer. The stereotyping decomposition below undercuts that reading: the LLM’s predictions concentrate on country identity, and the moderate rank correlations that do emerge reflect between-country differences rather than within-country household variation, a mechanism we develop next.

4.1 More Household Information Does Not Diminish Country-Level Stereotyping

The most direct way to test whether the moderate correlations reflect household-level signal or country-level generalizations is to regress PMT consumption predictions derived from LLM-generated data on true consumption with country fixed effects:

$$\hat{c}_i^{\text{LLM}} = \beta_0 + \beta_{\text{within}} \cdot \text{consumption}_i + \gamma_c + \epsilon_i \quad (1)$$

where γ_c is a country fixed effect. Because both the outcome and the regressor are measured in per-capita consumption dollars per day, β_{within} admits a direct interpretation: within country, a one-dollar-per-day increase in a household’s true

Table 1: Country Stereotyping vs. Household-Level Signal

Level	Granularity	N	β_{within}	R^2_{country}	R^2_{HH}
L1	Coarsest	453	0.033 (0.013)	0.56	0.104
L2	Coarse	3,518	0.015 (0.004)	0.51	0.064
L3	Standard	142,176	0.011 (0.002)	0.51	0.066
L4	Rich	142,176	0.024 (0.004)	0.42	0.170

Notes: Standard errors are clustered at the country level. N counts cells at L1/L2 and household-persona observations at L3/L4 (a household with multiple eligible adults generates one persona per adult).

per-capita consumption is associated with a β_{within} -dollar-per-day change in the LLM’s predicted consumption for that household. The complementary quantity, R^2_{country} , is the share of variance in the LLM’s predictions attributable to country identity alone, and indexes the extent to which LLM outputs are determined by country identity independently of household characteristics.

Table 1 presents the results. Within-country slopes are significant and positive at all four granularity levels (all $p < 0.05$), but the magnitudes are small (0.011–0.033). At L3, for example, $\beta_{\text{within}} = 0.011$: an extra dollar per day of true household consumption moves the LLM’s prediction for that household by about a cent, holding country constant. The substantive point is that true household consumption has little predictive power for PMT predictions derived from LLM-generated data once country identity is controlled for. Country fixed effects alone explain 43–56% of variance in these predictions; the incremental R^2 from adding true consumption on top of country (R^2_{HH}) is at most 17 percentage points.

The next question is whether the stereotype is present at the point the LLM generates individual variables or only emerges after the PMT aggregates those variables into a consumption prediction. Table 2 runs the same between-country variance decomposition on the individual LLM-generated PMT input variables (Claude Sonnet 5); each cell reports the share of variance in the variable attributable to country identity alone. Variable-level inflation is partial at each level and moves in the same direction as the aggregate stereotype. Variable-level inflation is partial at each level and moves in the same direction as the aggregate stereotype. Every variable is inflated at L3. At L4 rich context pulls most of them back toward their survey shares; number of children falls below its own, and female members remain inflated. The variable-level pattern rules out a pure aggregation artifact. The PMT downstream carries an inflation that is already present in the LLM’s individual variable draws.

The core result is that country-level stereotyping does not diminish as prompts add detail specific to a household. Raw R^2_{country} in the LLM’s predictions is 0.56 at L1, 0.51 at L2, 0.51 at L3, and 0.42 at L4. We normalize by the between-country share of variance in true household consumption

Table 2: Between-country share of variance in observed vs. LLM-generated PMT input variables.

Variable	True data	LLM (L3)	LLM (L4)
Household size	0.24	0.53	0.31
Number of children	0.37	0.57	0.30
Adults 18–64	0.11	0.29	0.15
Male members	0.05	0.10	0.08
Female members	0.05	0.14	0.15

(R^2_{country} for true welfare, which is 0.12 at L3), which asks how much the LLM’s predictions concentrate on country compared to how much country actually explains welfare in the data. The resulting stereotyping multiple is 1.4 at L1, 2.8 at L2, 4.4 at L3, and 3.5 at L4: the LLM concentrates variance on country identity between three and a half and four and a half times as much at the richer prompt levels as true household welfare actually varies between countries.

Two mechanisms plausibly produce this behavior. The first is a smoothing-under-sparsity effect. The training corpus contains progressively fewer observations at finer conditional cells: material referring to, for example, Senegal as a whole is common in text, but material referring specifically to 45-year-old female-headed households in the Kolda region of Senegal is not. When the prompt specifies a coarser cell, the model conditions on a well-supported region of its training distribution; when the prompt specifies a finer cell, the model conditions on a region with little direct training support, and its best available estimate is the coarser conditional it does have evidence for, which is the country prior. Additional household detail changes the query without supplying evidence that would help the model answer the finer query. The second mechanism is a decoder-side effect. Chat models fine-tuned with reinforcement learning from human feedback are pushed toward completions that are plausible and consistent with the whole prompt, and the completion most consistent with a long persona is typically the country-modal one, because that is the value most commonly reported for the country in whatever training text the model has drawn on.

A replication on US households separates these theories. We rerun the L1–L4 pipeline on 10,000 ACS households, substituting state for country and household income per capita for consumption. Sparsity predicts that stereotyping should attenuate where the aggregate unit is densely evidenced in training text; decoder pressure, a property of how the model generates rather than of what it has read, predicts that it should persist. We find that stereotyping attenuates with more unit-specific prompt detail in the US context (Figure 3). The LLM’s stereotyping multiple reads 1.4, 2.8, 4.4, 3.5 across L1–L4 in the 17 countries but stays at or below 1.4 at every US level (0.8, 0.8, 1.4, 1.1). The same pattern appears when we decompose the LLM’s own prediction variance. Group identity holds 54%, 56%, 51% and 42% of that variance across L1–L4 in our 17 countries, and 62%, 8%, 2.6% and 2.2% in the US. The US model still responds to the household it is given: at L3 the head’s own descriptors account for 18% of its prediction variance, against 25% in

the 17 countries, so its low state share is not noise. The ratio of persona share to group share measures how much the model attends to the household relative to the group it belongs to, and it moves from 0.48 in the 17 countries to 7.0 in the US. The result matches the sparsity prediction.

Santurkar et al. (2023) document a related asymmetry in opinion polling: LLM responses assigned to demographic subgroups reflect the distribution of viewpoints in the training data rather than the distribution held by those subgroups. Our result extends the same phenomenon into a quantitative-prediction setting where the group-level bias is measurable against ground truth. Farrell et al. (2025) offer a structural reading of both patterns that we will lean on in the discussion: LLMs are cultural technologies whose function is to encode, retrieve, and reproduce what is collectively recorded about the world. Country-level poverty patterns are extensively documented in development reports, academic literature, and public discourse; individual household outcomes, driven by idiosyncratic shocks such as crop failures, illness episodes, or labor-market disruption, are not. An LLM asked to represent a Burkinabe household retrieves the collectively-recorded material about Burkina Faso in the aggregate, and richer prompts do not supply new grounded material tied to the specific household.

The rest of this section reads the performance numbers through this mechanism. The largest coverage gap in Section 4.2 occurs at L3, the same level at which the stereotyping multiple peaks, and the fine-tuning result in Section 5 is consistent with an intervention that corrects group-level marginals without teaching the model within-group discrimination.

4.2 Consequence for Targeting: Inserting LLMs Degrades Targeting Relative to Statistical Baselines

Rank correlation does not fully answer the deployment-relevant question: does the LLM-PMT (a PMT scored on the LLM’s synthetic survey responses) outperform the True-PMT (the same PMT scored on real inputs)? Table 3 shows that the answer is consistently no; Appendix Figure A1 plots the same numbers. The True-PMT uses only the same information available to the LLM at each level, ensuring that performance differences reflect the cost of LLM generation, not differences in available information.

The LLM route underperforms the True-PMT at every level, and the gap is positive in the majority of countries at every level. At L1, the coverage gap is 8 percentage points (40% vs 47%); at L2, 6 (41% vs 47%); at L3, 13 (51% vs 64%); at L4, 7 (66% vs 73%). This is what the stereotyping result in Section 4.1 predicts: the LLM’s synthetic responses concentrate on country-typical values, and running a PMT on those responses pulls its predictions toward the country prior and away from the household variation the prompt inputs would otherwise pick up. Country-level heterogeneity is substantial: LLM-PMT exclusion errors at L3 range from 16% to 92% across countries. Distributional reconstruction (Appendix Figure A3, illustrated for Senegal) shows the same pattern at L3: the LLM-PMT systematically overes-

Table 3: Targeting Performance by Granularity Level

Level	N countries	Rank corr.	Coverage (%)		Gap (pp)
			LLM-PMT	True-PMT	
L1: Coarsest	17	0.38	40	47	8
L2: Coarse	17	0.41	41	47	6
L3: Standard	17	0.53	51	64	13
L4: Rich	17	0.69	66	73	7

Notes: Rank corr. = mean Spearman correlation between LLM cell means and actual cell means. Coverage = % of poor households reached when targeting the poorest fraction of households equal to the country-specific poverty rate (World Bank PIP at \$2.15/day, 2017 PPP); see Section 3.4. LLM-PMT scores a PMT on the LLM-generated survey responses plus the real L_k inputs the prompt states. True-PMT scores the same estimator on the actual survey responses to the same questions plus the same real inputs, so both routes see the same feature set at each level. At L4 the True-PMT uses the standard L3 PMT survey variables together with the L4 rich-context features; an alternative L4 True-PMT restricted to the rich-context features reaches 60% coverage against the LLM-PMT’s 66%, reversing the sign of the gap, though that comparison blinds the supervised model to the survey variables the LLM is synthesizing. Gap = True-PMT coverage minus LLM-PMT coverage. Poor defined as <\$2.15/day (2017 PPP).

timates consumption and within-cell variance collapses, so many poor households are classified as non-poor outright.

The L3 gap is the largest, at 13pp. The L4 gap narrows to 7pp because the L4 prompt anchors the LLM to concrete household observables and reduces its fallback to country-typical answers, consistent with the sparsity mechanism identified above. L4 is not itself a deployment scenario: it requires supplying every household’s assets, housing, and other survey variables in advance, and any organization with that data on hand can score a PMT on it directly rather than pass it through an LLM.

4.3 Who Gets Excluded: Disparate Impact Under the LLM-PMT

The pooled exclusion numbers hide who bears the cost of routing targeting decisions through an LLM. Figure 4 decomposes L3 exclusion errors across 17 countries by demographic group and by depth of poverty, comparing the LLM-PMT to the True-PMT under the same country-specific budget.

LLM-PMT’s excess exclusion error does not clearly concentrate on a particular demographic group outside of large households. Instead, in absolute terms it is dispersed among all demographic groups we examine, falling between 7.3 and 10.4 percentage points in every category outside household size. Male-led households have 30.5% excess exclusion error (33.5% vs 43.8%), while female-led households have 23.0% (43.3% vs. 53.3%), though female-led households face the higher exclusion rate under both routes. In relative terms the difference in error rates gets higher as poverty increases; the quintile deepest in poverty faces 51.4% excess exclusion error (16.6% vs 25.2%) while the quintile closest to the poverty line faces 18.3% (39.8% vs 47.1%). Household size shows the clearest signal, where households

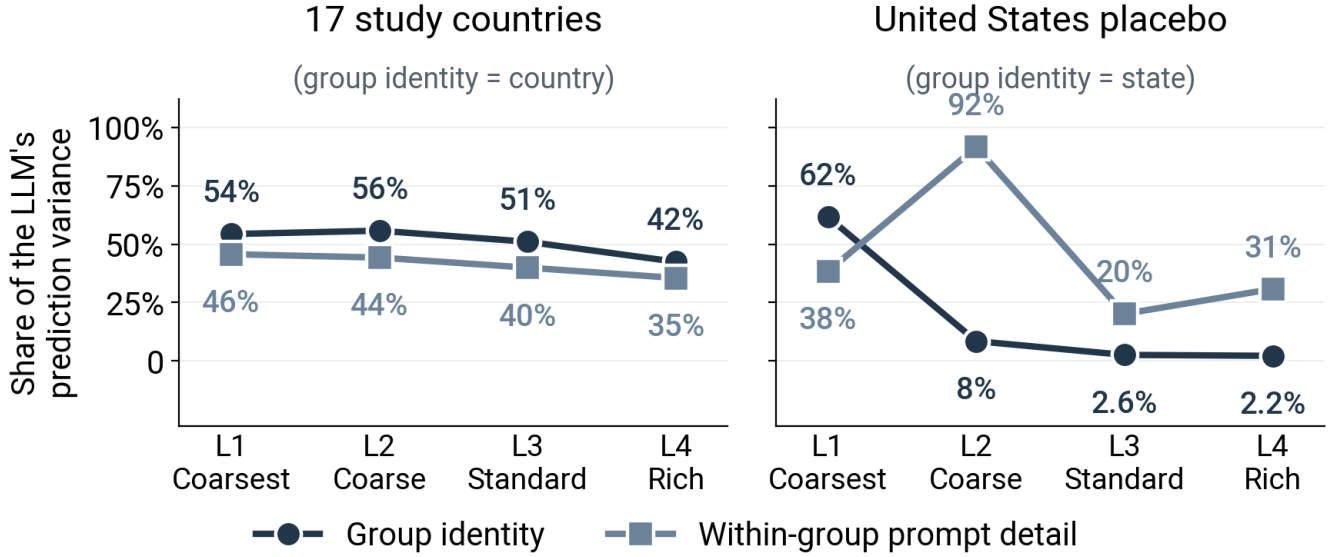


Figure 3: Share of the LLM’s prediction variance attributable to group identity and to within-group prompt detail, at each granularity level. Left: the 17 study countries, where the group is the country. Right: the US placebo, where the group is the state. Country identity holds close to half of the variance at every level; state identity falls away after L1.

of 1-2 people have lower exclusion errors using the LLM route by 32.9% (67.4% vs. 45.2%) while households of 10 or more people have 127.6% excess exclusion error (19.1% vs 43.5%). Routing through the LLM-PMT magnifies errors across the population and redistributes them from smaller households onto larger ones.

5 Fine-Tuning Improves Marginal Fit but Not Within-Cell Discrimination

One potential remediation for these excess errors is to fine-tune the model on empirical survey distributions, following Suh et al. (2025)’s approach in the opinion-polling setting. We do so, fine-tuning Qwen 2.5-7B-Instruct with LoRA (Hu et al. 2022) on matched question-response pairs from 14 country surveys, holding out three countries (Ghana, Malawi, Colombia) as fully out-of-sample tests. Training details are in Appendix D.

Fine-tuning replicates the distributional improvement Suh et al. document. Wasserstein distance between predicted and true consumption distributions falls in all 11 comparable countries by an average of 36%, and household-level Spearman rank correlation rises from a cross-country mean of 0.08 to 0.42 across the 10 evaluable countries. The three held-out countries also see gains (mean rank-correlation improvement of 0.23). Per-country slope charts across four metrics are in Appendix Figure A8.

The gains do not carry to targeting. The reason is informational. Fine-tuning on empirical survey distributions teaches the model the correct marginal distribution of each variable within each demographic cell, which is what improves distributional fit. It does not teach the model which specific households within a cell are poor, because within-

cell consumption depends on idiosyncratic events the training data does not observe. We confirm the cell-vs-household distinction directly on the held-out countries by comparing the fine-tuned model’s cell-level Spearman ρ (the upper bound for any constant-per-cell predictor) to the household-level ρ that results from broadcasting the cell-level prediction to every household in that cell. All three held-outs show the expected positive gap (Colombia +0.40, Malawi +0.14, Ghana +0.03; Appendix Figure A7). The within-cell heterogeneity that targeting requires is the part fine-tuning does not touch.

This design also allows us to answer a narrower question: not whether an LLM can or should replace or supplement surveys everywhere, but it can be relied upon to produce survey information for a country or region that has none. The held-out countries are that case. Marginals move toward the truth (mean rank-correlation gain of 0.23) while within-cell ordering does not, so a transferred model can support a poverty-rate estimate for the held-out country and still cannot rank the households inside it. That is the boundary of the useful case: under data sparsity, LLM transfer is a candidate tool for description and not for allocation, and it is allocation that L1–L4 evaluate. Where allocation is the goal and survey data are thin, the established alternatives are estimates built from satellite imagery and mobile phone records (Yeh et al. 2020; Chi et al. 2022). These are already deployed for targeting, and unlike an LLM’s output they are derived from observations of the population being targeted.

6 Discussion

The empirical results above establish that the LLM-PMT underperforms the True-PMT at every level of data richness we examine. We argue in this section that the underperformance

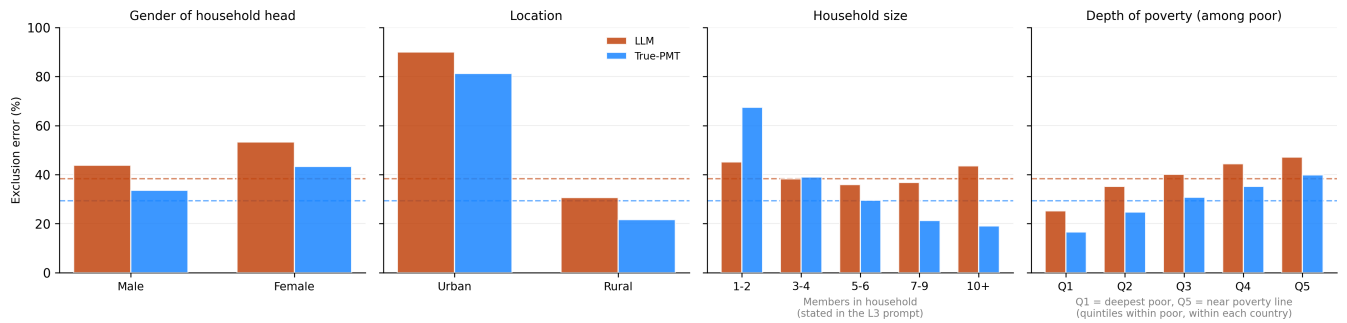


Figure 4: L3 exclusion errors by demographic group and depth of poverty, Claude Sonnet 5, pooled across 17 countries. Each bar shows the share of truly poor households in a group wrongly excluded from targeting when budgets are set to each country’s poverty rate. Dashed horizontal lines mark the pooled overall exclusion rate. Panels A and B break down by gender of head and urban/rural location; Panel C by household size, which the L3 prompt states; Panel D divides poor households into quintiles by depth of poverty within each country (Q1 = deepest poor, Q5 = closest to the line).

reflects an execution shortfall rather than missing information in the prompt, and that even were that shortfall to close, LLM-generated data would remain disqualifying for allocative decisions about specific households for reasons that do not depend on its current accuracy.

6.1 Information Constraints and Execution Shortfall

Section 4.1 identifies country stereotyping as the source of the LLM’s underperformance, and Section 4.3 documents its distributional consequences. A further diagnostic question is whether the LLM at least approaches what is achievable from the demographic inputs. The information-ceiling analysis reported in Figure 5 says the LLM underuses them.

After removing L2 cell fixed effects, actual observed PMT features explain a median of 37% of within-cell consumption variance across our 17 countries. This is the ceiling for methods using the current PMT feature set; the remaining 63% is idiosyncratic variance (crop failures, illness episodes, labor-market disruption) that no demographic-only method can access. At L3, the True-PMT approaches this ceiling with a median household-level ρ of 0.71 against an oracle upper bound of 0.77. Adding the LLM’s synthetic survey answers to the same PMT feature set pulls median ρ down to 0.57: the LLM’s contribution is negative relative to the demographic features alone, which sits alongside the coverage result in Section 4.2 and the stereotyping mechanism in Section 4.1.

One possible reading of the execution shortfall is that the LLM’s structured-generation constraint discards, or at least does not allow for the extraction of, information the model has internally. We test this with a linear ridge probe on Qwen 2.5-7B activations, adapting Jahanparast, Hong, and Chang (2026)’s activation-probing approach. Across four granularities, the probe’s household-level Spearman ρ is 0.15, 0.34, 0.32, and 0.42 at L1–L4, in every case at or below the LLM’s next-token performance and below the True-PMT. The activations do not carry within-cell signal the decoder is throwing away; the gap to the True-PMT is not an artifact of decoding but of the model’s representations themselves. This

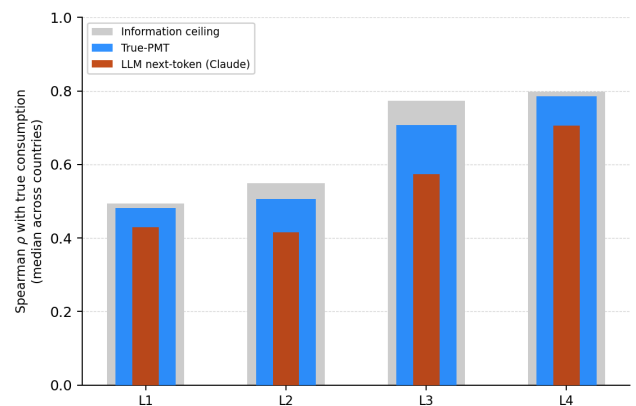


Figure 5: Median Spearman ρ with true per-capita consumption across 17 countries, by prompt granularity level. All bars are household-level. Gray: information ceiling (oracle upper bound). Blue: level-matched True-PMT. Rust: LLM (Claude) through next-token generation.

is consistent with the sparsity mechanism in Section 4.1: the internal representation of a specific-persona prompt has already collapsed to the country prior by the time the decoder reads it.

Fine-tuning tests the same limit from the training side (Section 5). The intervention works on its own terms: between-country concentration in the generated variables falls from above the survey’s own share to at or below it, and the coverage gap to a level-matched PMT closes. Targeting is unmoved. Exclusion error falls from 75% to 70% and improves in five of ten countries, because at L2 the fine-tuned model reaches a median household-level ρ of 0.45 against an information ceiling of 0.55, and a supervised PMT at that same level still excludes 53% of poor households. The binding constraint is the information in the prompt, not the capacity of the model reading it.

Together these results constrain the “wait for a better model” objection. Any between-cell knowledge a stronger

LLM might encode could be more efficiently supplied to a supervised model directly, since knowledge that can be characterized structurally belongs in features rather than in weights. And any within-cell signal such a model would need is not in the demographic inputs regardless of scale: no corpus contains a record of every household's specific shocks. This is not a claim that the current capability frontier is insufficient, it is a claim about the information set the LLM is being asked to condition on.

6.2 Recourse, Accountability, and the Sociotechnical Standards for LLM Substitution

Consider the simplest version of the targeting question: what is a legitimate reason to deny a household aid? The first-best answer is direct observation of household welfare. A welfare officer records assets, income, and consumption; the denial rests on facts about that specific household, which the household can verify, dispute, and if wrong, correct. When direct observation is too costly, second-best proxies are what programs in fact use. A PMT estimates welfare from observable indicators; a geographic poverty map ranks households by cell-level averages. Neither is ideal, but both preserve a partial connection to the first-best: denial still rests on something observable about the household or its immediate context.

The recourse point is the most direct. Eubanks (2018) and Benjamin (2023) document real failures of PMT-adjacent welfare systems: recorded inputs were wrong, model choices carried their own biases, and appeal processes were inaccessible in practice. Applied consistently, those critiques indict PMTs as well. PMTs are imperfect on all three values above. But a PMT is corrigible along a chain the household can act on, and an LLM-based imputation is not. When a PMT denial is challenged, the corrective step is a re-interview or re-verification of the specific input, and the fix is available to a household that has never trained a machine-learning model. The chain of reasoning terminates in a fact about the household that can be reproduced or repudiated. When an LLM-based denial is challenged, the corrective step is a change to the model or its training data, an intervention neither the household nor the deploying agency has purchase on, and one that would not address the specific case that generated the denial in any event. Contestability is a property of the sociotechnical infrastructure around a decision, not of the decision itself (Selbst et al. 2019), and that infrastructure requires a route from an individual challenge back to a change in the specific outcome. For PMTs that route exists and is imperfectly walked. For LLM-based imputation the route does not exist at all: the inference was never about the household, so there is nothing about the household to change. PMTs preserve a residual, actionable connection to the specific case that LLM-generated inputs abandon by construction.

Accountability tracks the same asymmetry. Development targeting is not the sociotechnical context that LLM survey simulation was developed for: opinion-polling errors aggregate across respondents and consequences are diffuse. Selbst et al. (2019)'s portability trap describes exactly the failure

mode of transplanting a tool between such contexts without accounting for how the new context changes what outputs mean and who bears the cost of errors. A decision-maker cannot readily be held responsible for an outcome whose inputs are unobservable to them and whose failure mode is a group-level statistical inference nobody can inspect.

Representativity connects the normative argument back to the mechanism. The problem is not group-level inference as such: a cell mean derived from actual survey responses within a country is a legitimate proxy, because it reflects something observed about households in that context. The LLM's group-level inference is categorically different. Farrell et al. (2025) characterize large AI models as cultural technologies that systematize and reproduce what is collectively recorded in text; for households in Burkina Faso or Ethiopia, that recorded material is produced predominantly in the Global North: development reports, academic literature, news coverage, policy documents. The LLM's representation of a Burkinabe household therefore reflects what that outside corpus encodes about Burkinabe households, not what the household itself would report, and this is what the stereotyping mechanism of Section 4.1 looks like from the normative side.

The failure mode generalizes beyond poverty targeting to other individual-case allocation settings (credit adjudication, healthcare triage, recidivism assessment); the same design (supervised baseline on the LLM's inputs, within-group fixed effect on true outcome) diagnoses it.

6.3 What a Defensible LLM Route Would Require

A larger model fine-tuned with a consumption-prediction head on curated multi-country welfare microdata or consumption diaries could plausibly close part of the empirical gap this paper documents. Two obstacles would remain: the normative objections of the previous subsection, and the absence of the training corpus such a model would need.

The normative objections developed in the previous subsection do not depend on empirical performance. A more capable model would still be trained on a corpus whose composition is undisclosed, whose contents cannot be audited by the households the model represents, and whose outputs cannot be traced to any specific observation about any specific household. Survey data have their own measurement problems: Jerven (2013) documents how African national accounts are estimated, revised, and disputed, and Beegle et al. (2012) shows experimentally that recall period and diary design substantially shift measured consumption in Tanzania. The difference lies in how each system responds to such problems. Measurement issues in survey data are identified, replicated, and argued about in the open, and an eligible respondent can in principle challenge the specific observation their eligibility was based on. The LLM system offers no comparable mechanism at either the aggregate or the individual level, regardless of model capability.

The training corpus a defensible LLM would require also does not exist and is unlikely to be produced. A curated multi-country diary-panel corpus of the size and richness needed to train a domain LLM would cost on the order of

hundreds of millions of dollars per round to collect. Development banks and national statistical offices are the only funders with budgets of that scale, and they have no incentive to produce a corpus whose primary application would be training a substitute for the surveys they themselves field. No other actor has a reason to pay for it. The dataset is a public good with no natural producer.

The ambient web-scraped alternative develops in the opposite direction. Shumailov et al. (2024) show that as pre-training data absorbs model-generated content, the tail of the empirical distribution collapses first, which is where poverty targeting draws its signal. Survey infrastructure produces new observations continuously, because someone walks to households and records their consumption; the LLM corpus does not, and in the absence of an intentional producer it degrades.

7 Limitations

Several limitations constrain the scope of our conclusions. Our evaluation uses Claude Sonnet 5 as the primary model; Qwen 2.5-7B serves as a robustness check but represents a much smaller model. It is reasonable to ask whether an even stronger model would close the shortfall we document, and Section 6.1 argues that it would not: a frontier model can reproduce group-level marginals more accurately, but it cannot recover within-group discrimination from inputs that do not contain it, and our stereotyping result shows that richer prompts do not move the model away from the country prior. Our targeting metrics also take consumption-ranked eligibility as the objective, which is the objective PMT-based programs are built to optimize; readers who reject that objective should read our comparisons as internal to it (Section 6.2).

Our PMTs are trained on 2,500 households per country, a sample size chosen to represent realistic deployment constraints. PMT performance would improve with larger training samples, but so would the individual-accuracy argument for using a PMT directly over LLMs.

The 17 countries in our sample represent predominantly sub-Saharan African and South Asian contexts; the generalizability of our findings to other regions with different survey data availability and LLM training data coverage is unclear.

8 Conclusion

We have provided evidence that LLM-generated survey responses concentrate on country-typical values rather than discriminating among households within a country. At L3, country identity explains 0.51 of the variance in the LLM’s downstream consumption predictions against 0.12 in true household consumption, a stereotyping multiple of 4.4; the LLM organizes 53% of its response variance around country and 25% around household persona. Consequences follow at the targeting stage: L3 exclusion errors from the LLM-PMT range from 16% to 92% across countries (mean 49%), versus 10% to 70% (mean 36%) for the level-matched True-PMT, and adding LLM-generated answers to a demographic PMT pulls median household-level rank correlation from 0.71 to 0.57. Fine-tuning reduces distributional divergence

by 36% on average across all 11 comparable countries, improves household-level rank correlation in all 10 evaluable countries, and lowers the country stereotyping to at or below the level in the surveys themselves. Exclusion error falls five percentage points, and in only five of those countries, because the fine-tuned model is already close to the information ceiling of its inputs. These technical failures are compounded by a structural epistemological problem: LLM-generated household data represents inference about demographic groups, not observation of specific households, creating an evidentiary gap that is inappropriate for high-stakes allocation decisions.

The development community faces genuine data constraints, and the search for lower-cost alternatives to comprehensive household surveys is understandable. But the adequacy of a measurement method must be evaluated against the requirements of its proposed application, not against cruder baselines. For targeting decisions that determine whether a poor household receives assistance, LLM-generated data falls short by a substantial margin at every level of granularity. Where data collection is feasible, the appropriate response is better surveys, not cheaper proxies.

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